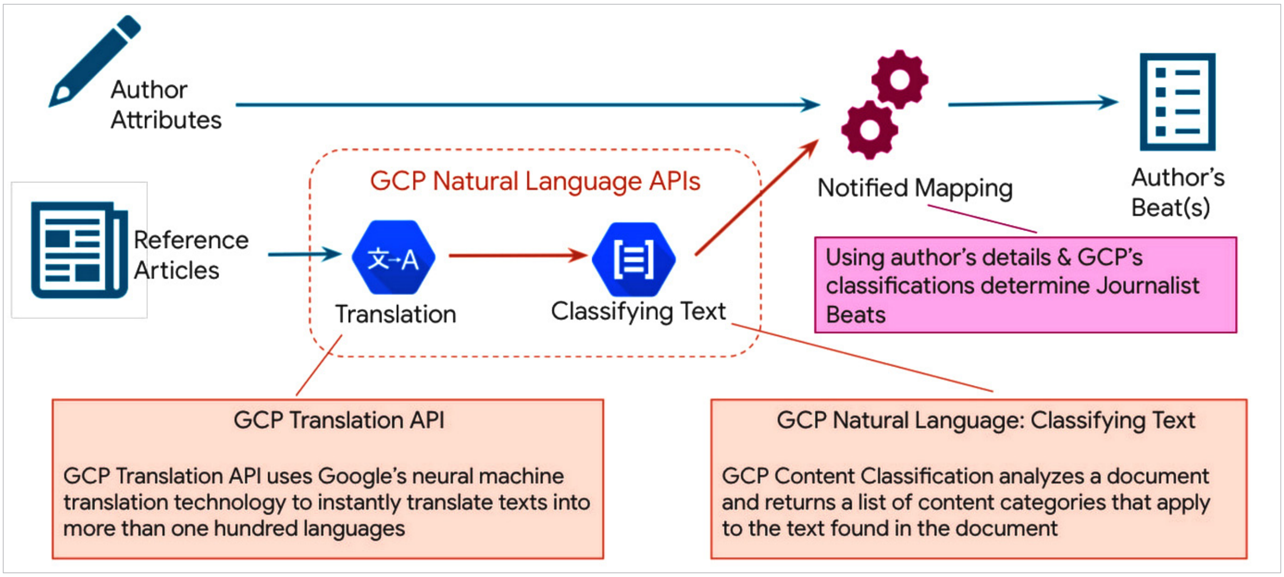

Figure 5: Google Cloud Natural Language API

Sentiment analysis and language translation can be achieved using the translation API services. This API helps with language detection, text extraction, language translation and sentiment analysis. Some common use cases are subtitle preparation for streaming videos, transcribing in multiple languages for video analytics, and Contact Center AI services for chatbots for local language support for banks and insurance company customers.

The Natural Language API in the GCP platform is powered by AutoML. It uses Vertex AI services for custom model preparation, and training, testing and deploying ML models for handling any NLU based use cases.

Google AutoML Vision

Automated machine learning (AutoML) is the process of applying machine learning to real-world problems in an automated manner. It covers the complete pipeline from the raw data set to the deployable machine learning model. It allows AI professionals to train customised models with specific data. It uses NAS (neural architecture search) to find the best ways to train these models. The only thing required is to gather data to enhance the model accuracy.

The two important stages of AutoML are described below.

Feature extraction: In the real world, machine learning needs tons of complex data to train models. Feature extraction is part of the dimensionality reduction process, i.e., raw data is divided and reduced to more manageable groups. These data sets have a large number of variables, which require strong computing power to process them. Feature extraction plays a crucial role here, as it reduces the number of resources without losing any critical content. It also reduces redundancy in the data.

Hyperparameter optimisation: All ML models have parameters that impact their quality. Without automated technology like hyperparameter tuning (applying AutoML), manual adjustments need to be made over the course of many trainings to arrive at the optimal hyperparameter values. Hyperparameter tuning makes the process of determining the best hyperparameter settings easier and less tedious.

Figure 6 highlights the working of AutoML.

AutoML Vision is a Google service for leveraging machine learning capabilities to classify images into thousands of predefined categories. It is useful when you have images you need to analyse intelligently, such as photos from security feeds. AutoML Vision operates in the Google Cloud and can be used either based on a graphical user interface, via REST, command line or Python. It implements strategies from NAS, currently a big subject of deep learning research. NAS is based on the idea that another model, typically a

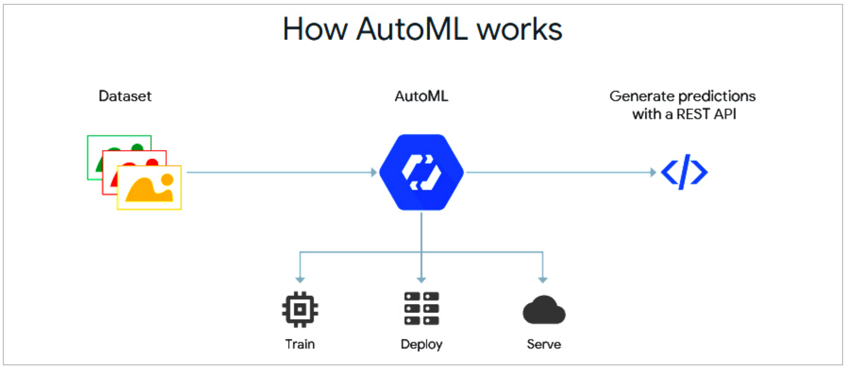

Figure 6: Google AutoML working